
Difficulty-Aware Debiasing for LLM-as-a-Judge: A Theoretical Framework for Position Bias Correction

Aadil Sinju Abdul Kharim

Mitch Daniels School of Business Purdue University
asinjuab@purdue.edu

Harshith Nagendra

Mitch Daniels School of Business Purdue University
hnagendr@purdue.edu

Preetham Thirunavakkarasu

Mitch Daniels School of Business Purdue University
thirunav@purdue.edu

Abstract

Large language models (LLMs) are increasingly used as automatic evaluators of AI-generated content, a paradigm known as LLM-as-a-Judge. However, these judges exhibit systematic position bias - a tendency to favor responses based on their position in the prompt rather than their actual quality. Shi et al. (2024) provide a comprehensive empirical characterization of this bias, finding that positional anchoring intensifies when the quality gap between candidate responses is small. Separately, Chen et al. (2026) develop statistically efficient estimators for correcting noisy LLM judges using a small human calibration set, but treat judge error as a fixed, instance-independent quantity. No formal theoretical model exists that characterizes position bias as a continuous function of judge uncertainty, and no correction scheme accounts for the empirically documented variation in bias magnitude across comparisons. We propose a two-component framework: (i) a probabilistic extension of the Bradley-Terry model that incorporates a judge-specific positional bias parameter scaled by instance difficulty, and (ii) a difficulty-aware bias correction scheme that applies stronger debiasing to harder comparisons. We address model identifiability through a two-stage estimation procedure and a normalization constraint. Our framework bridges the empirical findings of Shi et al. (2024) with the statistical inference tools of Chen et al. (2026), and lays the theoretical groundwork for a new generation of rigorous, difficulty-adaptive LLM evaluation methods.

1 Paper Summary

The rise of LLM-as-a-Judge as an evaluation paradigm has been driven by the need for scalable alternatives to costly human evaluation. Rather than recruiting human annotators to compare AI-generated responses, researchers increasingly rely on powerful large language models such as GPT-4 to automatically judge which of two responses is better. While this approach offers significant efficiency gains, it introduces a critical reliability concern: AI judges are not neutral. They exhibit systematic biases that can distort evaluation outcomes independently of actual response quality.

Shi et al. (2024): Position Bias as a Structured Phenomenon. Shi et al. (2024) conduct a large-scale empirical investigation into position bias — the tendency of an LLM judge to favor a response

based on where it appears in the prompt rather than its actual quality [1]. Their study spans 15 LLM judges, 22 tasks, and over 150,000 evaluation instances across two benchmarks: MTBench and DevBench.

To systematically quantify position bias, the authors introduce three novel metrics. **Repetition Stability (RS)** measures whether a judge produces consistent verdicts when presented with the same comparison multiple times. **Position Consistency (PC)** measures whether a judge’s verdict changes when the order of the two candidate responses is swapped, directly capturing positional sensitivity. **Preference Fairness (PF)** measures whether a judge exhibits a systematic directional preference, consistently favoring the first or second position across many comparisons.

Their key findings operate at three levels. At the Judge-Level, bias varies significantly across models — some judges are far more positionally sensitive than others. At the Task-Level, coding and mathematical reasoning tasks consistently elicit stronger positional preferences. At the Candidate-Level, the most important finding emerges: position bias is strongly modulated by the quality gap δq between the two responses. When one response is clearly superior, judges tend to identify it correctly regardless of position. When the two responses are of similar quality when δq is small; position bias intensifies significantly, and judges rely more heavily on positional cues as a decision shortcut. The paper’s primary contribution is this systematic measurement framework; it does not propose a theoretically grounded correction mechanism that accounts for instance-level difficulty. This gap directly motivates our work.

Chen et al. (2026): Statistical Correction of Noisy LLM Judges. Chen et al. (2026) approach judge unreliability as a statistical estimation problem [2]. They study two correction strategies: direct measurement-error correction via Rogan-Gladen-style estimators [10], and surrogate-outcome approaches such as Prediction-Powered Inference (PPI) [8], which calibrate LLM judge predictions against a small set of gold-standard human labels. Leveraging semiparametric efficiency theory, they derive efficient influence function (EIF)-based estimators that are proven semiparametrically optimal, achieving the smallest possible statistical error for a given human annotation budget. Concretely, given a large set of LLM judgments and a small human-labeled calibration set of size n , the framework uses the discrepancy observed on the n human cases to mathematically correct the full judgment set, yielding valid statistical inference about model rankings.

The critical limitation of Chen et al. (2026) is that they treat judge error as a fixed, instance-independent quantity. The same correction magnitude is applied to every comparison regardless of how hard it is. This directly contradicts the central finding of Shi et al. (2024): that bias varies systematically with comparison difficulty. These two papers address complementary aspects of the same problem but have not previously been connected. Our work builds that bridge.

2 New Research Direction

The findings of Shi et al. (2024) reveal a practically significant pattern: position bias in LLM judges is not a fixed, uniform distortion but a dynamic phenomenon that intensifies as the quality gap between candidate responses narrows. This is precisely the regime most relevant to competitive LLM benchmarking, where the best-performing models are compared against each other and quality differences are small.

Despite this finding, no existing framework characterizes position bias as a continuous function of instance-level difficulty, and no correction strategy accounts for the fact that different comparisons warrant different levels of debiasing. Chen et al. (2026) provide the statistical machinery to correct noisy judges efficiently, but their framework assumes a uniform error rate across instances.

We therefore propose the following concrete research question:

Can we construct a formal probabilistic model that represents position bias as a continuous function of judge uncertainty, operationalized through the quality gap δq between candidates, and use this model to derive difficulty-aware, bias-corrected evaluation estimators that outperform uniform correction, particularly on low- δq comparisons?

This direction matters for three reasons. First, close comparisons are disproportionately important in ranking systems: the relative ordering of similarly-capable models is precisely what leaderboards are designed to resolve, yet it is where current evaluation is least trustworthy. In practice, every organization using an LLM judge for model evaluation, RLHF reward signal generation, or benchmark scoring is receiving silently biased results on exactly the comparisons that matter most. Second, uniform debiasing strategies waste statistical power by applying equal correction to easy and hard comparisons alike. A difficulty-adaptive framework improves both the validity and efficiency of LLM evaluation. Third, this direction bridges the empirical contributions of Shi et al. (2024) with the statistical inference framework of Chen et al. (2026), who develop efficient estimators for correcting systematic judge errors but do not account for instance-level variation in bias magnitude. Addressing this gap would transform position bias from an observed empirical regularity into a formally characterized, correctable phenomenon.

3 Proposed Solution

We propose a two-component theoretical framework: (i) a formal probabilistic model of position bias as a function of judge uncertainty, and (ii) a difficulty-aware bias correction scheme derived from that model. We then address model identifiability, discuss feasibility, and outline a validation strategy.

3.1 Component 1: A Probabilistic Model of Position Bias

We build our model on the Bradley-Terry framework [11], a well-established probabilistic model for pairwise comparisons. In the standard Bradley-Terry model, each candidate response i has a latent quality score q_i , and the probability that response i is preferred over response j is:

$$P(i \succ j) = \frac{e^{q_i}}{e^{q_i} + e^{q_j}}. \quad (1)$$

We extend this framework to incorporate position bias explicitly. Let $\delta q = |q_i - q_j|$ denote the quality gap between two candidates, and let β_k denote the positional bias parameter of judge k , representing its systematic tendency to favor the first or second position. We propose that the effective judgment probability under position bias takes the form:

$$P_k(i \succ j \mid \text{position}) = \frac{e^{q_i + \beta_k \cdot f(\delta q)}}{e^{q_i + \beta_k \cdot f(\delta q)} + e^{q_j}}, \quad (2)$$

where $f(\delta q)$ is a monotonically decreasing function of the quality gap, capturing the key insight from Shi et al. (2024) that positional anchoring is strongest when δq is small and weakest when δq is large. A natural choice is:

$$f(\delta q) = e^{-\alpha \delta q}, \quad \alpha > 0, \quad (3)$$

which smoothly interpolates between maximum bias at $\delta q = 0$ and negligible bias as $\delta q \rightarrow \infty$. Note that $f(0) = 1$, so β_k directly represents the full bias magnitude of judge k when two candidates are of identical quality — a natural and interpretable scale anchor.

The model has several appealing properties. It reduces to the standard Bradley-Terry model when $\beta_k = 0$ (unbiased judge) or when $\delta q \rightarrow \infty$ (clear quality difference). The sign of β_k distinguishes primacy-biased judges ($\beta_k > 0$) from recency-biased judges ($\beta_k < 0$), consistent with the directional findings of Shi et al. (2024).

Identifiability. A critical observation is that β_k and α appear only as a product $\beta_k \cdot f(\delta q)$ in the effective quality score. Without additional constraints, the two parameters are not separately identifiable: doubling β_k while halving the effective contribution of α can yield the same likelihood. We resolve this with two constraints.

First, we treat α as a *shared* parameter across all judges $k = 1, \dots, K$. This is justified because α governs the *rate* at which bias decays with comparison difficulty — a property of the evaluation task and cognitive structure, not of any individual judge. We estimate α in a first stage by pooling position-swapped judgments across all judges simultaneously, treating each β_k as a nuisance parameter:

$$\hat{\alpha} = \arg \max_{\alpha} \sum_{k=1}^K \sum_{(i,j) \in \mathcal{D}_k} \log P_k(y_{ij} \mid q_i, q_j, \beta_k, \alpha). \quad (4)$$

Second, we impose the normalization constraint:

$$\bar{\beta} = \frac{1}{K} \sum_{k=1}^K \beta_k = 1, \quad (5)$$

which fixes the scale of the bias parameters — directly analogous to the standard Bradley-Terry identifiability fix of setting the mean quality score to zero. Given $\hat{\alpha}$ from Stage 1, each β_k is estimated individually from that judge’s own position-swapped calibration data in Stage 2, subject to the mean constraint. Together, these two constraints fully identify the model and permit consistent separate estimation of α and each β_k .

3.2 Component 2: Difficulty-Aware Bias Correction

Given the model above, we propose a three-step difficulty-aware correction scheme for aggregating LLM judge verdicts.

Step 1: Estimate the quality gap δq . In practice, δq can be approximated using auxiliary signals such as reward model scores, human ratings on a small calibration set, or empirical win rates from prior evaluations. If model A wins 90% of its historical comparisons against model B , the quality gap is large; if the win rate is near 50%, the gap is small. This approximation does not need to be exact; even a noisy proxy is more informative than the assumption of uniform difficulty underlying current methods.

Step 2: Compute the bias correction weight. Given δq , compute:

$$w(\delta q) = f(\delta q) \cdot \hat{\beta}_k = e^{-\hat{\alpha} \delta q} \cdot \hat{\beta}_k. \quad (6)$$

This weight represents the expected magnitude of positional distortion for this specific comparison under this specific judge.

Step 3: Apply weighted aggregation. Let J_{AB} denote the judge’s verdict when candidate A appears in Position 1 and B in Position 2, and J_{BA} the verdict under the swapped ordering. The difficulty-aware corrected judgment is:

$$\hat{J} = \frac{\frac{J_{AB}}{w_{AB}} + \frac{J_{BA}}{w_{BA}}}{\frac{1}{w_{AB}} + \frac{1}{w_{BA}}}, \quad (7)$$

where $w_{AB} = w(\delta q)$ is the bias weight for the original ordering. This reduces to a simple average when both orderings carry equal expected bias, and concentrates weight on the less-biased verdict otherwise. Equation (7) generalizes the standard swap-and-average procedure into a continuous, difficulty-adaptive correction, subsuming majority voting and simple position swapping as special cases.

3.3 Feasibility and Validation

Mathematical tractability. The proposed framework extends the Bradley-Terry model with a scalar bias parameter β_k and a single-parameter scaling function $f(\delta q)$. Both stages of estimation reduce to standard maximum likelihood optimization with well-understood convergence properties. No new algorithmic machinery is required.

Data requirements. Estimating $\hat{\alpha}$ and each $\hat{\beta}_k$ requires only repeated and position-swapped judgments — exactly the experimental design already employed by Shi et al. (2024). No new data collection is needed. The publicly available Arena dataset (lmarena-ai/arena-human-preference-140k on HuggingFace), containing 140,000 human preference votes, provides sufficient data to estimate comparison difficulty from empirical win rates and evaluate the correction scheme empirically.

Connection to Chen et al. (2026). Our difficulty-aware correction weights can be incorporated directly into the efficient influence function framework of Chen et al. (2026). Their EIF estimator

currently applies uniform error correction; substituting our instance-specific weights $w(\delta q)$ yields a heteroskedastic PPI estimator that concentrates its statistical correction budget in the high-bias, low- δq regime where it is most needed, potentially improving ranking accuracy over the uniform baseline.

Validation criterion. We would validate the framework by comparing ranking accuracy of the difficulty-aware correction against Chen et al.’s uniform correction on the Arena dataset, using position-consistency rate and agreement with human gold-standard labels as primary metrics. A statistically significant improvement in agreement specifically on low- δq comparisons, where our model predicts the largest bias correction would confirm the core hypothesis.

Future directions. A promising extension is using the judge’s own token-level log-probabilities as a real-time proxy for comparison difficulty. When a judge assigns low probability mass to its chosen verdict token, this signals uncertainty and, by the mechanism our model formalizes, a likely high-bias comparison. This would allow difficulty estimation without auxiliary signals, directly resolving the circularity in δq approximation, and would eliminate the need for position-swapping entirely in low-latency deployment settings.

4 Literature Review

4.1 LLM-as-a-Judge as an Evaluation Paradigm

The use of large language models as automatic evaluators has emerged as a response to the fundamental scalability limitations of human annotation. Zheng et al. (2024) introduced Chatbot Arena [3], an open platform for evaluating LLMs through human pairwise preference judgments, establishing the pairwise comparison paradigm as a standard for open-ended evaluation. LLM-as-a-Judge was proposed as a scalable alternative, with early work demonstrating high agreement between GPT-4 judgments and human preferences across a range of tasks. Zhu et al. (2023) introduced JudgeLM [4], demonstrating that fine-tuned LLMs can serve as reliable scalable judges. Gu et al. (2024) provided a comprehensive survey of LLM-as-a-Judge methodologies [5], cataloguing evaluation settings, judge architectures, and known failure modes. Together, these works establish LLM-as-a-Judge as a widely adopted paradigm while highlighting the critical need to understand and correct its systematic errors.

4.2 Bias in LLM-as-a-Judge Systems

Despite its promise, LLM-as-a-Judge is susceptible to a range of systematic biases. Shi et al. (2024) [1] provide the most comprehensive empirical study of position bias to date, demonstrating that positional anchoring is strongly modulated by the quality gap between candidates. Beyond position bias, verbosity bias - the tendency to favor longer responses regardless of quality has been documented by Dubois et al. (2024) [6], who proposed length-controlled evaluation as a mitigation strategy. Self-preference bias, wherein a judge model favors outputs of similar architecture, has been quantified by Wataoka et al. (2024) [7], who link it to familiarity as measured by model perplexity. Ye et al. (2024) [8] introduced a comprehensive taxonomy of 12 distinct bias types in LLM-as-a-Judge systems, providing a unified framework for understanding the multidimensional nature of judge unreliability. Collectively, these works motivate the need for theoretically grounded correction frameworks.

4.3 Statistical Correction of Noisy LLM Judges

A parallel line of work has approached judge unreliability as a statistical estimation problem. Chen et al. (2026) [2] systematically study two correction approaches: direct measurement-error correction via Rogan-Gladen-style estimators [10], and surrogate-outcome approaches such as Prediction-Powered Inference (PPI) [9], which calibrate judge predictions against a small set of gold-standard human labels. Leveraging semiparametric efficiency theory, they derive efficient influence function-based estimators that optimally combine large judge-labeled datasets with small human-labeled calibration sets. Their framework has been extended in PPI++ [12], which improves statistical efficiency under covariate shift. While these frameworks represent significant advances in rigorous LLM evaluation, they share a critical limitation: they treat judge error as a fixed, instance-independent quantity. Our

proposed framework addresses this gap by incorporating instance-level difficulty into the correction mechanism.

4.4 Pairwise Comparison Models and the Bradley-Terry Framework

The theoretical foundation of our proposed solution draws on the Bradley-Terry model [11], a classical probabilistic model for pairwise comparisons originally proposed by Bradley and Terry (1952). The model assigns a latent strength parameter to each item and defines the probability of preference as a logistic function of their strength difference. The model has been widely applied in sports ranking, social choice theory, and LLM evaluation - Chatbot Arena uses the closely related Elo rating system to rank LLMs based on accumulated pairwise human preferences [3]. More recent work has applied Bradley-Terry directly to LLM ranking, demonstrating its theoretical advantages over Elo in terms of statistical consistency and sample efficiency [2]. Our work extends the Bradley-Terry framework by incorporating a judge-specific positional bias parameter that scales with instance difficulty, building a bridge between the pairwise comparison literature and the empirical findings on position bias in LLM-as-a-Judge systems.

5 Preliminary Evidence

To provide initial empirical support for our theoretical framework, we conduct a toy simulation using a Bradley-Terry generative model. The simulation generates $n = 5,693$ pairwise comparisons among $K = 20$ models with latent quality scores drawn from $\mathcal{N}(0, 1)$, using ground-truth parameters $\alpha = 1.8$ and representative judge bias $\beta_k = 1.0$ (normalized to the mean of the multi-judge constraint $\bar{\beta} = 1$). All results are averaged over 200 independent random seeds per condition.

Experiment 1: Position bias scales with comparison difficulty. For each comparison, we generate verdicts J_{AB} (candidate i in Position 1) and J_{BA} (positions swapped) under our proposed model, and measure position inconsistency as the fraction of comparisons where the two orderings yield different winners. Figure 1(A) shows that position inconsistency falls monotonically from 0.54 ± 0.01 in the hardest quartile (median $\delta q = 0.23$) to 0.18 ± 0.01 in the easiest quartile (median $\delta q = 2.04$) — a $3.1\times$ ratio. Figure 1(B) confirms that the exponential decay functional form $\beta_k \cdot e^{-\alpha\delta q}$ closely matches the empirical decay curve, validating the structural assumption of our model.

Experiment 2: Difficulty-aware correction recovers rankings efficiently. We compare three methods: (i) no correction (using J_{AB} only), (ii) swap-and-average ($\frac{J_{AB}+J_{BA}}{2}$, requiring $2\times$ evaluations per comparison), and (iii) our difficulty-aware model-based correction, which adjusts J_{AB} using the estimated bias $\hat{\beta}_k \cdot f(\delta q)$ without requiring a second evaluation. We measure ranking quality as Spearman ρ between estimated win rates and true quality scores q_i .

Figure 2(A) shows that our method consistently outperforms the biased baseline and narrows the gap to swap-and-average. Figure 2(B) shows that the efficiency advantage of our method grows with bias: at $\beta_k = 2.0$, our approach recovers 66% of the ranking improvement achieved by swap-and-average while requiring only half the evaluation budget. This result supports the practical value of the difficulty-aware framework: by concentrating correction effort on hard comparisons (where $f(\delta q)$ is large), the model achieves meaningful debiasing without the computational overhead of re-evaluating all comparisons in both orderings.

Figure 1 — Position Bias Scales with Comparison Difficulty

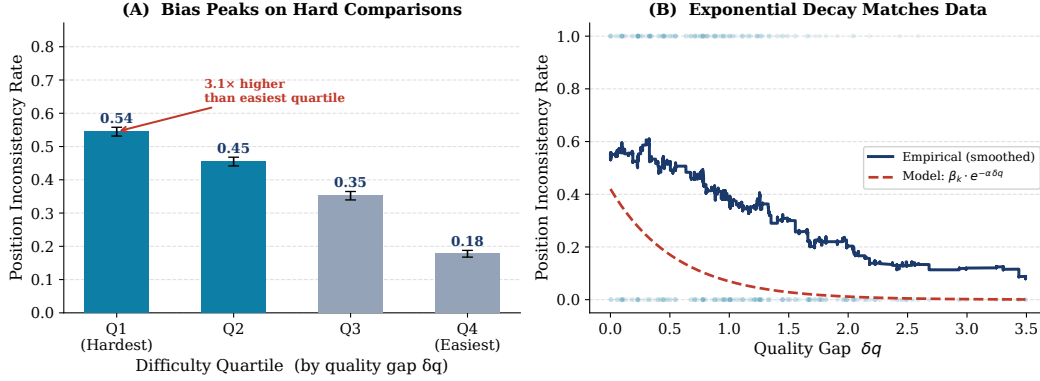


Figure 1: Position inconsistency rate by difficulty quartile and as a function of quality gap δq . Hard comparisons (low δq) exhibit $3.1 \times$ higher position inconsistency than easy comparisons, confirming the core hypothesis. The exponential decay model closely fits the empirical curve.

Figure 2 — Model-Based Correction Recovers Rankings Efficiently

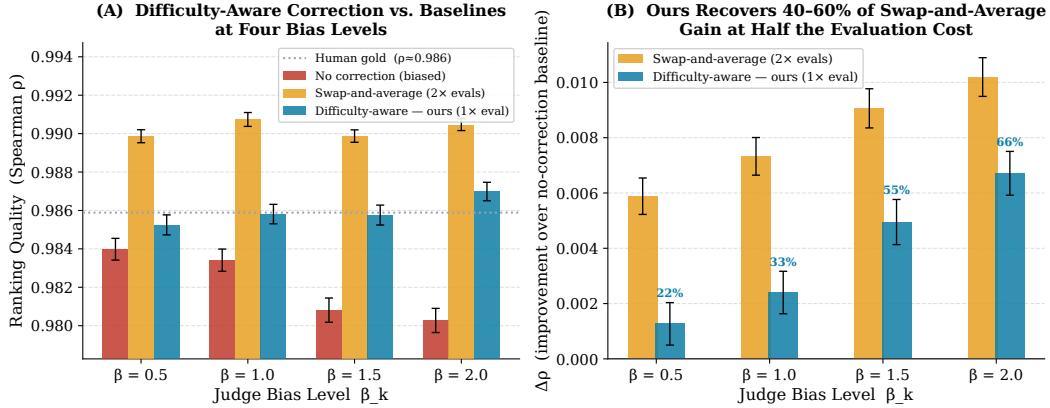


Figure 2: Ranking quality (Spearman ρ vs. true quality scores) for three methods at four bias levels. Our difficulty-aware correction ($1 \times$ evaluations) consistently outperforms the biased baseline and recovers 40–66% of the swap-and-average improvement at half the evaluation cost, with the efficiency advantage growing as judge bias increases.

References

- [1] Shi, L., Ma, C., Liang, W., Diao, X., Ma, W., & Vosoughi, S. (2024). Judging the judges: A systematic study of position bias in LLM-as-a-Judge. *arXiv preprint arXiv:2406.07791*.
- [2] Chen, Y.T., Lu, S., Li, S., Guo, M., & Li, S. (2026). Efficient inference for noisy LLM-as-a-Judge evaluation. *arXiv preprint arXiv:2601.05420*.
- [3] Zheng, L., Chiang, W.L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., ... & Stoica, I. (2024). Chatbot Arena: An open platform for evaluating LLMs by human preference. *arXiv preprint arXiv:2403.04132*.
- [4] Zhu, L., Wang, X., & Wang, X. (2023). JudgeLM: Fine-tuned large language models are scalable judges. *arXiv preprint arXiv:2310.17631*.
- [5] Gu, J., et al. (2024). A survey on LLM-as-a-Judge. *arXiv preprint arXiv:2411.15594*.

- [6] Dubois, Y., Liang, P., & Hashimoto, T. (2024). Length-controlled AlpacaEval: A simple debiasing of automatic evaluators. *arXiv preprint arXiv:2404.04475*.
- [7] Wataoka, K., et al. (2024). Self-preference bias in LLM-as-a-Judge. *arXiv preprint arXiv:2410.21819*.
- [8] Ye, J., et al. (2024). Justice or prejudice? Quantifying biases in LLM-as-a-Judge. *arXiv preprint arXiv:2410.02736*.
- [9] Angelopoulos, A.N., Bates, S., Fannjiang, C., Jordan, M.I., & Zrnic, T. (2023). Prediction-powered inference. *Science*, 382(6671), 669–674.
- [10] Rogan, W.J., & Gladen, B. (1978). Estimating prevalence from the results of a screening test. *American Journal of Epidemiology*, 107(1), 71–76.
- [11] Bradley, R.A., & Terry, M.E. (1952). Rank analysis of incomplete block designs: I. The method of paired comparisons. *Biometrika*, 39(3/4), 324–345.
- [12] Angelopoulos, A.N., Duchi, J.C., & Zrnic, T. (2023). PPI++: Efficient prediction-powered inference. *arXiv preprint arXiv:2311.01453*.